Date

DAY- 11 (After mid 1)

Chapter 3

Concept of random process/variables

→ A random variable is a function that associates a real number with each element in the sample space.

→ we will use capital letters to denote variable

→ we will use small letters for one of its values.

A discrete random variable assumes each of its values with a certain probability.

DAY-12

We shall denote by $f(x), g(x) \cdots$ where x is

$f(x) = P(X = x)$; that is $f(3) = P(X = 3)$

the set of ordered pairs $(x, f(x))$ is called probability distribution.

$$\left.\begin{array}{l} \rightarrow f(x) \geq 0 \\ \rightarrow \sum_x f(x) = 1 \\ \rightarrow P(X = x) = f(x) \end{array}\right\} \text{three basic condition to be a probability distribution}$$

Writing $F(x) = P(X \leq x)$, we define $F(x)$ to be cummulative distribution function of a random variable X.

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \text{ for } -\infty < x < \infty$$

the graph only increases or stays constant but never go down.

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the function $f(x)$ is probability density function (pdf) for a continuous random variable X, if

$\left.\begin{array}{l} \rightarrow f(x) \geq 0, \text{ for all } x \in R \\ \rightarrow \int_{-\infty}^{\infty} f(x)\,dx = 1 \\ \rightarrow P(a < X < b) = \int_a^b f(x)\,dx \end{array}\right\}$ three conditions